

Course- B.Tech.	Type- Core
Course Code- 2025CSET106	Course Name- Discrete Mathematical Structures
Year- I	Semester- Even (2 nd Semester)

Tutorial-13

Tutorial No.	Name	CO1	CO 2	CO3
13	Addition modulo m, Multi. Modulo m, Ring, Field, Poset, Lattice	✓	--	✓

Objective: The main objective of this tutorial is to help students take the foundational concepts of discrete math including modular arithmetic, algebraic structures, POSETs, and graph theory and actually apply them to solve practical problems.

Q1. (a). Check if following groups are cyclic.

- (i) $(\{1, \omega, \omega^2\}, \times)$
- (ii) $(\{1, -1, i, -i\}, \times)$
- (iii) $(\mathbb{Z}_4, +)$

(b). Prove that the set of integers $\{0, 1, 2, 3, 4, 5, 6\}$ forms a Ring under $+_7$ and $\times_7$ operations.

Q2. For a set $A = \{1, 2, 3\}$, check if the following relations are POSETs?

$$R1 = \{(1,1), (2,2), (3,3)\},$$

$$R2 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$$

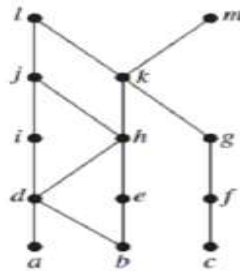
$$R3 = \{ \}$$

Q3. Consider the set $A = \{4, 5, 6, 7\}$. Let R be the relation $\leq$ on A . Draw the directed graph and the Hasse diagram of R .

Q4. Let $X = \{1, 2, 3, 4, 5, 6\}$. Draw Hasse diagram for $(X, |)$, where $|$ represents the 'divides' relation.

Q5. For the Hasse diagram given below, find maximal, minimal, greatest, least, elements of the given Hasse diagram. Also find lower bounds, upper bounds, greatest lower bound, least upper bound for the following subsets.

- a) $\{d, k, f\}$
- b) $\{b, h, f\}$
- c) $\{d\}$
- d) $\{a, b, c\}$
- e) $\{l, m\}$



Q6. Determine whether the POSETs represented by each of the Hasse diagrams are lattices?

